Theory Questions

Stored Procedure

11. What is a stored procedure?
12. Why do we use stored procedures?
13. Difference between procedure and function?
14. Can a stored procedure return values?
15. What are input and output parameters?
16. How do you create a stored procedure?
17. How do you execute a stored procedure?
18. Can a procedure call another procedure?
19. What are the advantages of stored procedures?
20. What are disadvantages of stored procedures?

◆ **Trigger**

21. What is a trigger?
22. When are triggers executed?
23. Types of triggers in SQL?
24. Difference between BEFORE and AFTER trigger?
25. What is an INSTEAD OF trigger?
26. Can a trigger call a procedure?
27. What are the limitations of triggers?
28. How do you create a trigger?
29. How do you disable or drop a trigger?
30. What are real-time use cases of triggers?

◆ **Function**

31. What is a function in SQL?
32. Difference between function and stored procedure?
33. Types of functions in SQL?
34. What is a scalar function?
35. What is a table-valued function?
36. Can a function return multiple values?

37. Can we use DML statements inside a function?

38. How do you create a function?

39. What do you call a function?

40. What are the advantages of functions?

◆ **Cursor**

41. What is a cursor?

42. Why do we use cursors?

43. Types of cursors?

44. What are the steps to use a cursor?

45. Difference between cursor and loop?

46. What are implicit and explicit cursors?

47. What are cursor attributes?

48. What is a forward-only cursor?

49. What are the disadvantages of cursors?

50. How do you close and deallocate a cursor?

◆ **ACID Properties**

51. What are ACID properties?

52. What is Atomicity?

53. What is Consistency?

54. What is Isolation?

55. What is Durability?

56. What is a transaction?

57. What is the isolation level?

58. Types of isolation levels in SQL?

59. What are dirty reads, non-repeatable reads, phantom reads?

60. How does ACID ensure data integrity?

◆ **Miscellaneous Questions**

61. What is normalization?

62. What are different normal forms?

63. What is denormalization?

64. What is a primary key and foreign key?

65. What is a composite key?

66. What is indexing and how does it improves performance?

67. What is a constraint in SQL?

68. Difference between DELETE, TRUNCATE, and DROP?

69. What is a transaction log?

70. What is database locking?

◆ **Scenario-Based Questions**

71. When would you use a cursor instead of a set-based query?

72. When should you avoid triggers?

73. How would you improve performance of a stored procedure?

74. How do you handle errors in stored procedures?

75. How would you design a system ensuring ACID compliance?

MCQ Questions

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1. What is a stored procedure?

A. Table
B. Precompiled SQL code
C. Index
D. View

Answer: B
Explanation: Stored procedure is precompiled SQL code stored in DB.

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2. Why do we use stored procedures?

A. Reduce code reuse
B. Improve performance
C. Delete tables
D. Create indexes

Answer: B
Explanation: Precompiled → faster execution.

---

3. Difference between procedure and function?

A. Function returns value
B. Procedure returns value

C. Both same
D. None

Answer: A

---

4. Can a stored procedure return values?

A. No
B. Yes
C. Only string
D. Only integer

Answer: B
Explanation: Using OUT parameters.

---

5. What are input parameters?

A. Output values
B. Passed into procedure
C. Return values
D. None

Answer: B

---

6. Which command is used to execute procedure?

A. CALL
B. RUN
C. EXEC
D. Both A & C

Answer: D

---

7. Can a procedure call another procedure?

A. No
B. Yes
C. Only once
D. Only in function

Answer: B

---

8. Advantage of stored procedure?

A. Slow execution
B. Security

C. No reuse
D. None

Answer: B

---

9. Disadvantage of stored procedure?

A. Easy debugging
B. Hard to maintain
C. Fast execution
D. Secure

Answer: B

---

10. What is a trigger?

A. Function
B. Automatic event
C. Manual query
D. Index

Answer: B

---

11. When are triggers executed?

A. Before/After event
B. Only after select
C. Manual execution
D. Never

Answer: A

---

12. Types of triggers?

A. BEFORE, AFTER
B. INSERT only
C. DELETE only
D. None

Answer: A

---

13. BEFORE trigger runs?

A. After operation
B. Before operation
C. During select
D. Never

Answer: B

---

14. INSTEAD OF trigger is used?

A. Replace operation
B. Delete data
C. Create table
D. None

Answer: A

---

15. Can trigger call procedure?

A. No
B. Yes
C. Only once
D. Only in MySQL

Answer: B

---

16. Limitation of triggers?

A. Easy debugging
B. Complex logic
C. Fast
D. None

Answer: B

---

17. What is a function in SQL?

A. Returns value
B. Table
C. Index
D. Cursor

Answer: A

---

18. Scalar function returns?

A. Table
B. Single value
C. Multiple values
D. None

Answer: B

19. Table-valued function returns?

A. Single value
B. Table
C. Index
D. Null

Answer: B

---

20. Can function return multiple values?

A. No
B. Yes (table)
C. Only integer
D. Only string

Answer: B

---

21. What is a cursor?

A. Table
B. Row-by-row processor
C. Index
D. Function

Answer: B

---

22. Why use cursor?

A. Bulk processing
B. Row-by-row processing
C. Delete table
D. Indexing

Answer: B

---

23. Types of cursor?

A. Static, Dynamic
B. Only static
C. Only dynamic
D. None

Answer: A

---

24. Cursor disadvantage?

A. Fast
B. Slow
C. Secure
D. None

Answer: B

---

25. What is Atomicity?

A. All or nothing
B. Partial update
C. Slow
D. None

Answer: A

---

26. What is Consistency?

A. Data integrity
B. Speed
C. Index
D. None

Answer: A

---

27. What is Isolation?

A. Transactions independent
B. Merge
C. Delete
D. None

Answer: A

---

28. What is Durability?

A. Temporary data
B. Permanent save
C. Delete
D. None

Answer: B

---

29. What is normalization?

A. Remove redundancy
B. Add redundancy
C. Delete data
D. None

Answer: A

---

30. Difference between DELETE and TRUNCATE?

A. DELETE faster
B. TRUNCATE faster
C. Same
D. None

Answer: B
Explanation: TRUNCATE removes all rows quickly.

Quantitative Aptitude

Aptitude topics MCQ Question

Time & Work

1. A can complete a work in 10 days, B in 15 days. Together they take?

A. 5 days

B. 6 days

C. 7 days

D. 8 days

Answer: B

Solution:

A = 1/10, B = 1/15

Total = 1/10 + 1/15 = (3+2)/30 = 5/30 = 1/6

Time = 6 days

2. A does a job in 12 days, B in 18 days. How long together?

A. 6

B. 7.2

C. 8

D. 9

Answer: B

Solution:

1/12 + 1/18 = (3+2)/36 = 5/36

Time = 36/5 = 7.2 days

3. A is twice as efficient as B. If B takes 12 days, A takes?

A. 4

B. 5

C. 6

D. 8

Answer: C

Solution:

Efficiency ratio = 2:1

Time ratio = 1:2

A = 12/2 = 6 days

4. A can do work in 20 days. After 5 days, B joins and completes in 10 more days. B alone takes?

A. 15

B. 20

C. 25

D. 30

Answer: B

Solution:

A does 5/20 = 1/4

Remaining = 3/4

(A+B) does 3/4 in 10 days → rate = 3/40

$A = 1/20$

$B = (3/40 - 1/20) = 1/40 \rightarrow$ Time = 40 days

Correction → Option mismatch → closest logical = 40 (not listed)

5. A, B, C complete work in 6, 8, 12 days. Together?

A. 2

B. 3

C. 4

D. 5

Answer: B

Solution:

$1/6 + 1/8 + 1/12 = (4+3+2)/24 = 9/24 = 3/8$

Time = $8/3 \approx 2.67 \approx 3$ days

6. If 4 men do work in 8 days, 8 men take?

A. 2

B. 4

C. 6

D. 8

Answer: B

Solution:

Work = constant

Men × Days = constant

$4 \times 8 = 32$

$8 \times x = 32 \rightarrow x = 4$ days

7. A finishes in 15 days. B is 50% more efficient. Time taken by B?

A. 8

B. 10

C. 12

D. 14

Answer: B

Solution:

Efficiency B = 1.5A

Time = 15 / 1.5 = 10 days

8. A can do in 9 days, B in 18 days. Working alternately starting A → total days?

A. 10

B. 11

C. 12

D. 13

Answer: C

Solution:

A+B in 2 days = 1/9 + 1/18 = 1/6

6 cycles = 12 days

9. Pipe fills tank in 6 hrs, leak empties in 12 hrs. Net time?

A. 10

B. 12

C. 8

D. 6

Answer: B

Solution:

1/6 - 1/12 = 1/12

Time = 12 hrs

10. 5 workers finish work in 20 days. 10 workers take?

A. 5

B. 10

C. 15

D. 20

Answer: B

Solution:

5×20 = 100

10×x = 100 → x = 10 days

Mixture & Alligation

11. Mix 10L water in 40L milk. Ratio?

A. 4:1

B. 1:4

C. 2:5

D. 5:2

Answer: A

Solution:

Milk = 40L, Water = 10L

Ratio (Milk : Water) = 40 : 10

Divide by 10 → 4 : 1

12. 30L solution has 20% alcohol. Alcohol?

A. 5

B. 6

C. 7

D. 8

Answer: B

Solution:

Alcohol = 20% of 30

= (20/100) × 30

= 6L

13. 10L milk, 20% replaced with water. Pure milk left?

A. 6

B. 7

C. 8

D. 9

Answer: C

Solution:

20% of 10L = 2L removed (pure milk)

Remaining milk = 10 − 2 = 8L

14. Ratio 3:2, total 25L. Milk?

A. 10

B. 12

C. 15

D. 18

Answer: C

Solution:

Total parts = 3 + 2 = 5

Milk = (3⁄5) × 25

= 15L

15. Mix 10L (₹20/L) and 10L (₹30/L). Avg price?

A. 22

B. 24

C. 25

D. 28

Answer: C

Solution:

Total cost = (10×20) + (10×30) = 200 + 300 = 500

Total quantity = 20L

Average price = 500 / 20 = ₹25/L

16. Alligation: 10% and 30% → 20%. Ratio?

A. 1:1

B. 2:1

C. 1:2

D. 3:2

Answer: A

Solution (Alligation Method):

Higher − Mean = 30 − 20 = 10

Mean − Lower = 20 − 10 = 10

Ratio = 10 : 10 = 1 : 1

17. Add water to reduce 50% to 25% in 10L. Water added?

A. 5

B. 10

C. 15

D. 20

Answer: B

Solution:

Initial milk = 50% of 10 = 5L

Let final volume = x

25% of x = 5

x = 20L

Water added = 20 − 10 = 10L

18. Ratio 2:3, total 50L. Water?

A. 20

B. 25

C. 30

D. 35

Answer: C

Solution:

Total parts = 2 + 3 = 5

Water = (3⁄5) × 50

= 30L

19. Replace 10% twice in 100L. Milk left?

A. 81

B. 80

C. 82

D. 79

Answer: A

Solution:

Formula: Remaining = Initial × (1 − x)n

Here, x = 10% = 0.1, n = 2

Milk left = 100 × (0.9)2

= 100 × 0.81

= 81L

20. Mix 20% and 40% → 30%. Ratio?

A. 1:1

B. 2:1

C. 1:2

D. 3:2

Answer: A

Solution (Alligation):

40 − 30 = 10

30 − 20 = 10

Ratio = 10 : 10 = 1 : 1

Permutations & Combinations

21. 5! = ?

A. 60

B. 100

C. 120

D. 150

Answer: C

Solution:

Factorial means multiplication of all positive integers up to that number:

$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

22. Number of ways to arrange 3 letters?

A. 3

B. 6

C. 9

D. 12

Answer: B

Solution:

Arrangement of 3 distinct letters = 3!

$= 3 \times 2 \times 1 = 6$ ways

23. 5P2 = ?

A. 10

B. 20

C. 25

D. 30

Answer: B

Solution:

Formula: $nPr = n! / (n-r)!$

$5P2 = 5! / 3!$

= (5 × 4 × 3!) / 3!

= 5 × 4 = 20

24. 5C2 = ?

A. 10

B. 20

C. 5

D. 15

Answer: A

Solution:

Formula: nCr = n! / [r!(n−r)!]

5C2 = 5! / (2! × 3!)

= (5 × 4 × 3!) / (2 × 1 × 3!)

= 20 / 2 = 10

25. Arrange "ABC"?

A. 3

B. 6

C. 9

D. 12

Answer: B

Solution:

All letters are distinct

Total arrangements = 3!

= 3 × 2 × 1 = 6

26. 6C3 = ?

A. 15

B. 20

C. 30

D. 60

Answer: B

Solution:

6C3 = 6! / (3! × 3!)

= (6 × 5 × 4 × 3!) / (3! × 3!)

= (6 × 5 × 4) / (3 × 2 × 1)

= 120 / 6 = 20

27. Ways to choose 2 from 4?

A. 4

B. 5

C. 6

D. 8

Answer: C

Solution:

4C2 = 4! / (2! × 2!)

= (4 × 3 × 2!) / (2 × 1 × 2!)

= 12 / 2 = 6

28. Arrange 4 objects → ?

A. 12

B. 16

C. 24

D. 32

Answer: C

Solution:

Arrangement of 4 distinct objects = 4!

= 4 × 3 × 2 × 1 = 24

29. 4P2 = ?

A. 8

B. 10

C. 12

D. 16

Answer: C

Solution:

4P2 = 4! / (4−2)!

= 4! / 2!

= (4 × 3 × 2!) / 2!

= 4 × 3 = 12

30. Ways to select 3 from 5?

A. 10

B. 15

C. 20

D. 25

Answer: A

Solution:

5C3 = 5! / (3! × 2!)

= (5 × 4 × 3!) / (3! × 2 × 1)

= 20 / 2 = 10

Coding Questions

🔷 Stored Procedure

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1. Fetch employees with salary greater than input

Description:
You need to design a procedure that accepts a salary value and returns all employees earning more than that amount.

Example:
Employees → [5000, 8000, 12000]

Input → 7000
Output → Employees with 8000, 12000

---

2. Insert record and return ID

Description:
Create a procedure that inserts a new employee and returns the generated ID.

Example:
Insert → "John, 10000"
Output → New ID (e.g., 101)

---

3. Update salary by department

Description:
Procedure should update salaries of all employees in a specific department.

Example:
Dept 10 → Increase salary by 10%
All employees in Dept 10 get updated salaries

---

4. Add two numbers

Description:
Procedure takes two inputs and returns their sum using an output parameter.

Example:
Input → 10, 20
Output → 30

---

5. Delete old records & log

Description:
Delete records older than a specific date and store them in a log table.

Example:
Delete → records before 2020
Log → store deleted data for backup

🔷 Trigger

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6. Prevent negative salary

Description:
Trigger checks salary before inserting and blocks if negative.

Example:
Insert → salary = -5000
Result → Rejected

---

7. Audit salary update

Description:
Whenever salary changes, store old and new values.

Example:
Old → 5000
New → 7000
Audit table → (5000 → 7000)

---

8. Update last_modified column

Description:
Automatically update timestamp whenever record is updated.

Example:
Update name → last_modified updated automatically

---

9. INSTEAD OF trigger on view

Description:
Insert into a view, but trigger redirects data to actual tables.

Example:
Insert into view → data goes to Employee + Department tables

---

10. Restrict deletion

Description:
Prevent deletion if record is active.

Example:
Active employee → cannot delete

◈ Function

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11. Factorial function

Description:
Return factorial of a number.

Example:
Input → 5
Output → 120

---

12. Total salary by department

Description:
Return sum of salaries for a department.

Example:
Dept 10 → salaries [5000, 7000]
Output → 12000

---

13. Employees with salary > value

Description:
Return table of employees whose salary is above input.

Example:
Input → 6000
Output → employees > 6000

---

14. Nth highest salary

Description:
Find nth highest salary from table.

Example:
Salaries → [1000,2000,3000]
n=2 → 2000

---

15. Format full name

Description:
Combine first name and last name.

Example:
Input → John, Doe
Output → John Doe

◈ **Cursor**

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16. Increase salary by 10%

Description:
Process each employee row one by one and update salary.

Example:
5000 → 5500

---

17. Conditional insert using cursor

Description:
Fetch records and insert only those meeting condition.

Example:
Salary > 5000 → insert into another table

---

18. Department-wise salary

Description:
Calculate total salary for each department using cursor.

Example:
Dept 10 → 10000
Dept 20 → 15000

---

19. Update row by row

Description:
Update each record individually based on condition.

Example:
If salary < 5000 → increase

---

20. Print employee names

Description:
Cursor fetches each name and prints sequentially.

Example:
John
Smith
Alex

💠 ACID & Transactions

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21. Money transfer

Description:
Transfer money between accounts safely.

Example:
A → 1000 → 900
B → 500 → 600
If failure → rollback

---

22. Multi-table insert

Description:
Insert into multiple tables, rollback if any fails.

Example:
Insert into Orders + Payments
Failure → rollback both

---

23. COMMIT & ROLLBACK

Description:
Save changes or undo them.

Example:
Insert data → commit = saved
Error → rollback = undone

---

24. Dirty read

Description:
One transaction reads uncommitted data.

Example:
Transaction A updates salary but not committed
Transaction B reads it

---

25. SERIALIZABLE isolation

Description:
Transactions behave as if executed sequentially.

Example:
No interference between transactions

🔷 Miscellaneous

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26. Table with constraints

Description:
Create table with PK, FK, unique.

Example:
Employee ID (PK), Dept ID (FK), Email (Unique)

---

27. Index creation

Description:
Create index to speed up search.

Example:
Search by name → faster using index

---

28. Normalize to 2NF

Description:
Remove partial dependency.

Example:
Split table into smaller related tables

---

29. Composite key

Description:
Primary key made of multiple columns.

Example:
(StudentID + CourseID)

---

30. DELETE vs TRUNCATE vs DROP

Description:

- DELETE → removes rows
- TRUNCATE → removes all quickly
- DROP → deletes table

Example:
DELETE → remove some rows
TRUNCATE → empty table
DROP → remove table